DEVELOPING A CENTRALIZED GRADING PORTAL WITH AUTOMATED TRANSMUTATION AND PERFORMANCE RISK ALERTS TO ENHANCE GRADE SUBMISSION AND STUDENT ACADEMIC MONITORING IN ARELLANO UNIVERSITY - ELISA ESGUERRA CAMPUS

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Background of the Study

At Arellano University - Elisa Esguerra Campus, the way student records are handled is still very traditional. It involves a lot of manual data entry and physically moving grade sheets from one office to another. Every time grading season hits, teachers are buried in paperwork that must be signed and passed around—starting from the subject instructors, then to the senior faculty for review, and finally to the principal for approval. This kind of manual headache is why many schools are switching to automated systems to handle their organizational gradebooks (Classter, 2026). Since all this information is spread across different folders and documents, it is difficult for a teacher to get a quick look at how a student is actually doing. This makes it far too easy to miss someone who has missing assignments or is starting to fall behind.

The need for a digital upgrade has become even more urgent because of recent DepEd policy changes. Moving toward a "no grade transmutation" policy means grades now have to reflect a student's actual raw score without any of the traditional padding. While that makes sense for honesty, it takes away a "safety net" many students previously relied on to pass. Now, the pressure is on teachers to be 100% accurate with calculations and to keep a much closer eye on progress. Using digital grading features like those found in openSIS (n.d.) can help manage these strict requirements while keeping the data accurate. Without a digital tool to flag struggling students automatically, a failing status might not be realized until the semester is basically over. Recent research shows that using natural language processing and automated data can make this grading process much more precise (Ayaan & Ng, 2025).

This research advocates for the development of a Centralized Grading Portal specifically for the Elisa Esguerra Campus. The goal is to swap out the messy paper-based system for an automated digital workflow that calculates final averages directly from raw scores while staying fully compliant with the new DepEd rules. Building this kind of automated system is a proven way to modernize how grades are tracked (Araujo & Kim, n.d.). By including a "Performance Risk Alert" feature, the portal will act as an early warning, letting teachers know exactly who is falling behind or missing work long before the final deadlines. This matches what experts say about "academic performance warning systems" being the best way to use data to help students in higher education (Duong et al., 2022). In the end, this initiative is about making the grading process smoother and making sure no student slips through the cracks because of a slow, manual system.

Statement of the Problem

The overall aim of this study is to develop a Centralized Grading Portal with Automated Transmutation and Performance Risk Alerts for Arellano University - Elisa Esguerra Campus. This project is designed to replace the manual record-handling process with an automated system that handles grade calculations and risk monitoring. The following specific and measurable objectives are derived from this aim and are set to be achieved within the current academic semester:

Specifically, this study seeks to answer the following questions:

1. What is the current status of the grading and monitoring process at Arellano University - Elisa Esguerra Campus in terms of manual data entry, compliance with DepEd's no-transmutation policy, and identification of at-risk students?
2. How can a Centralized Grading Portal be designed and developed to automate grade transmutation and provide real-time performance risk alerts?
3. How effective is the developed portal as perceived by the faculty and administrators in terms of:
    3.1. Accuracy of grade calculations;
    3.2. Efficiency of the submission workflow; and
    3.3. Promptness of academic monitoring and risk identification?

Methodology

Research Design

This study uses a **Quantitative-Developmental** research design. The main goal is to develop the grading portal and then test its performance using objective data. Instead of using surveys to gather opinions, the researchers will focus on measuring the system's accuracy and how much time it takes to process grades compared to the manual method. This approach ensures that the portal is technically sound and meets the needs of the school.

Locale and Subject of the Study

The study will be conducted at Arellano University - Elisa Esguerra Campus. This location was chosen because the researchers observed that the faculty still uses a manual, paper-based process for submitting grades and tracking student progress. The subject of the study will be the grading records and templates used by the campus. To protect privacy, the researchers will use mock data or anonymized records during the testing phase of the portal.

Technical Framework

The development of the Centralized Grading Portal is built on the idea that technology should actually make a teacher's life easier rather than adding another task to their to-do list. To explain why this digital shift is so important for our campus, this study is built on two main ideas: the Technology Acceptance Model (TAM) and Information Processing Theory.

First, the Technology Acceptance Model (Davis, 1989) tells us that for any new tool to work, the people using it, our teachers, need to feel that it is actually useful and not a headache to learn. Right now, the manual task of carrying grade sheets from office to office and doing all that math by hand is a huge burden. By moving to a Centralized Grading Portal, we are giving them a system that is useful because it handles the signatures and easy to use because everything is in one digital spot. If the portal saves them time and cuts down on stress, they will be much more likely to leave the old paper system behind.

Second, Information Processing Theory looks at how we handle a lot of data at once. At the Elisa Esguerra Campus, student info is currently scattered across different folders and papers, which makes it really hard for a teacher to see how everyone is doing at a glance. This theory explains that when we are overwhelmed with manual data entry, it is easy to make mistakes or miss a student who is starting to fail. The portal acts like an extra brain for the teacher. It does the heavy lifting of calculating raw scores and uses the Performance Risk Alert to automatically flag struggling students. This ensures we stay on top of the new "no-transmutation" rules and makes sure no student slips through the cracks just because the old manual system was too slow to keep up.

Conceptual Framework

The researchers followed a step-by-step process to build the portal, ensuring that all DepEd grading rules, like the "no-transmutation" policy, are properly coded into the system.

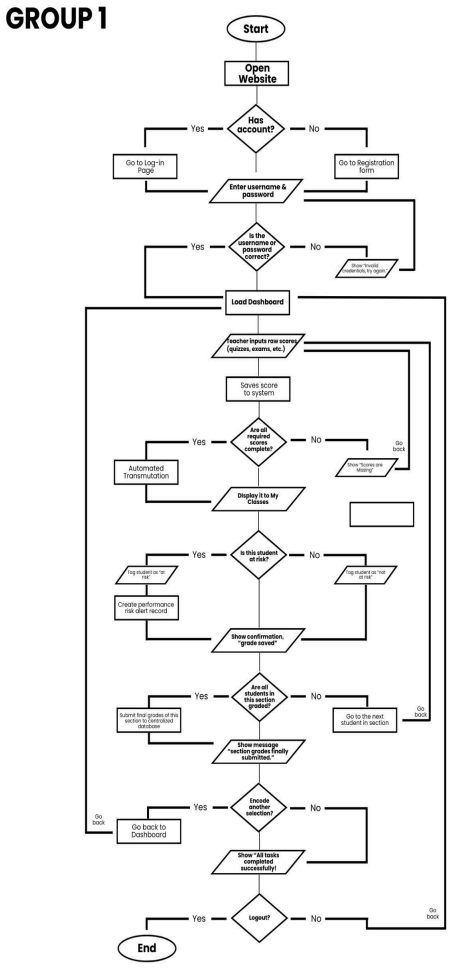

Figure 1. Research Flowchart of the Centralized Grading Portal

Data Collection and Testing Procedure

Because this study does not use a survey, the data will be collected through a series of system tests:

1. **Accuracy Testing:** The researchers will compute a set of grades manually and then process the same data through the portal. The results will be compared to ensure the portal's calculations are 100% accurate.
2. **Efficiency Testing:** The researchers will measure the time it takes to identify at-risk students manually versus the time it takes for the portal to generate an automated alert.

Statistical Treatment of Data

To prove the portal's effectiveness, the following simple statistics will be used:

- **Accuracy Rate:** A percentage used to show if the portal's math matches the manual calculations.
- **Time Differential:** A comparison used to show the amount of time saved by using the automated system instead of the manual workflow.

Significance of the Study

The goal of this project is to fix the manual grading problems at Arellano University - Elisa Esguerra Campus. The following groups will benefit from this study:

Faculty Members. The teachers are the ones most affected by the current manual system. This portal will make their work much easier by automating the grade calculations and transmutation. It removes the need for manual data entry and physical paperwork, allowing them to submit grades faster and with fewer errors.

School Administrators. This study will help the administration by providing a centralized way to monitor grades. It will be easier for them to track if teachers are following the new DepEd "no-transmutation" rules and to see the overall academic performance of the campus without having to look through physical folders.

Students. Even though they don't use the portal directly, students will benefit from the "Performance Risk Alert" feature. This allows their teachers to catch if a student is failing early on, giving the student more time to ask for help or catch up on missing assignments before the semester ends.

Future Researchers. Students who want to build similar school systems in the future can use this study as a guide. They can look at our process and our results to help them develop even better versions of automated academic tools.